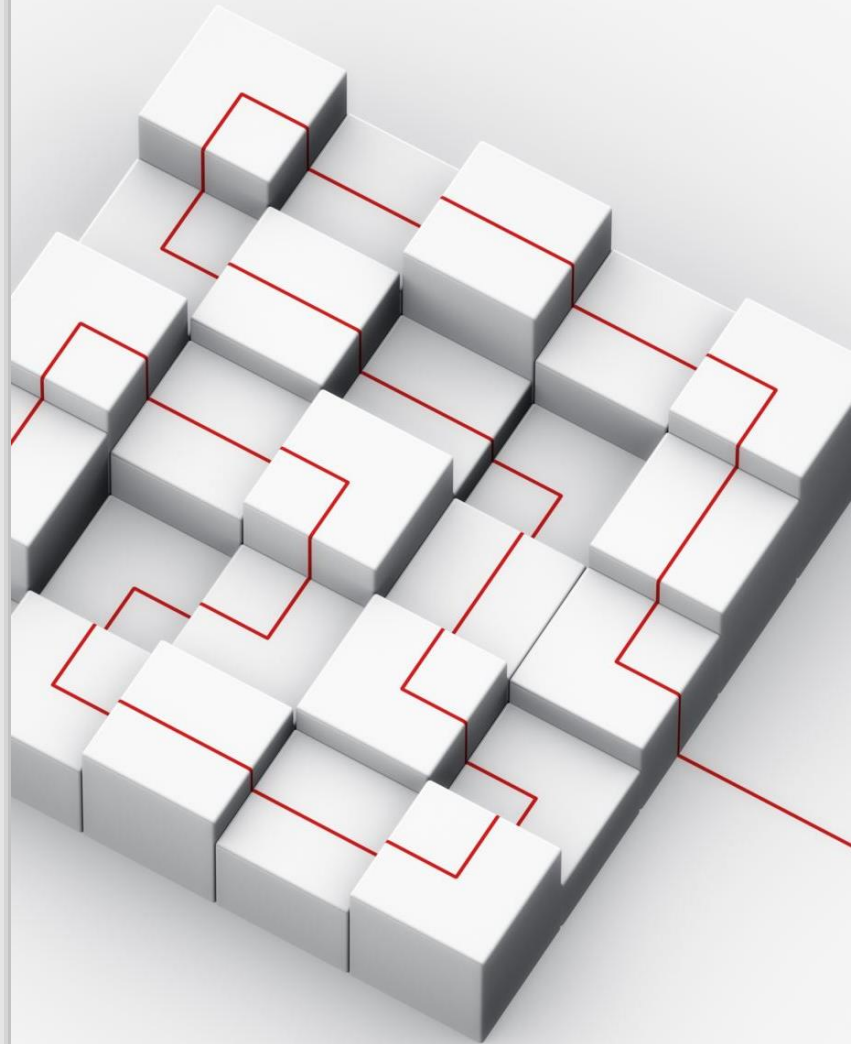




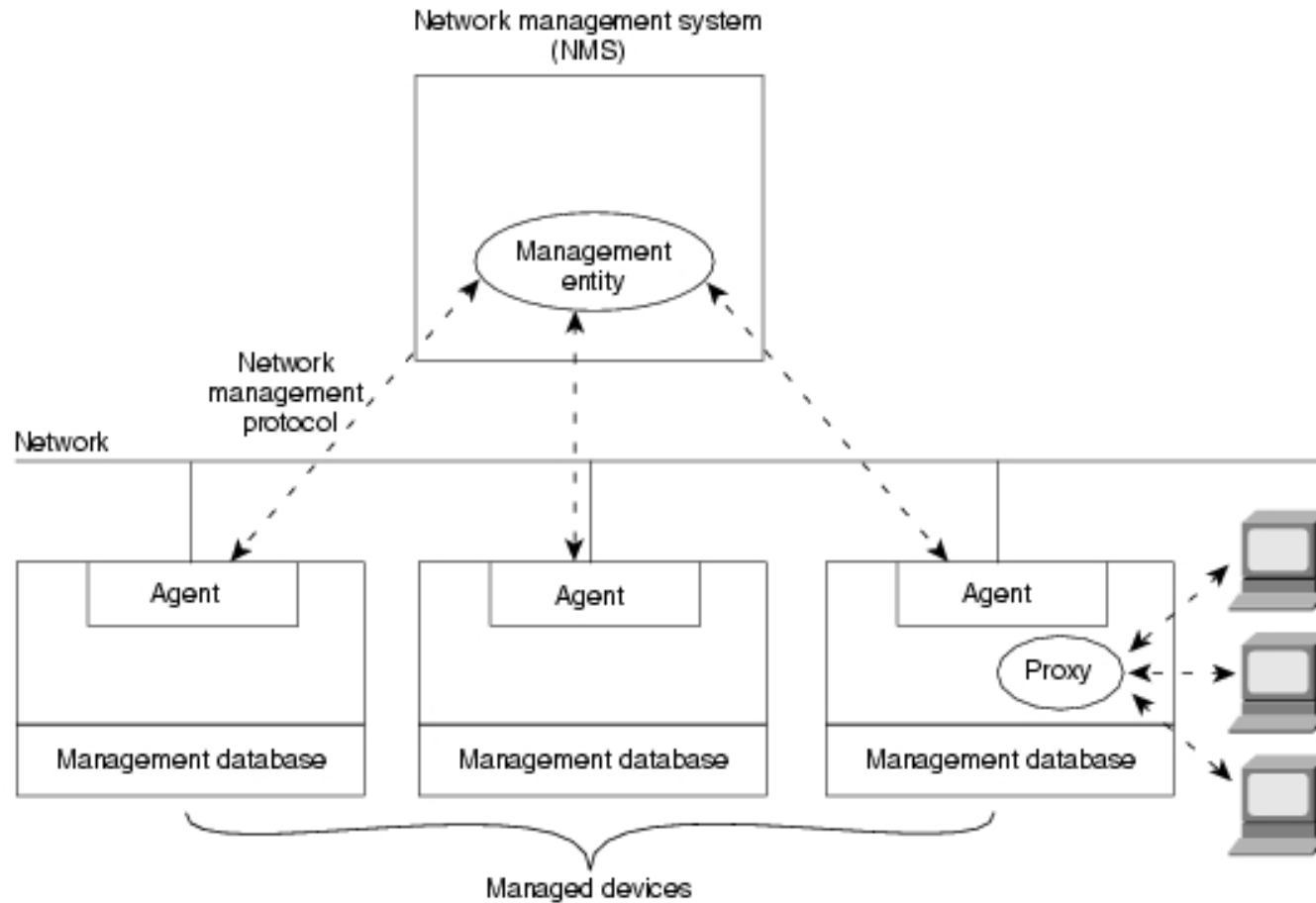
Network Management Architecture

Lesson Objectives

- *At the end of this topic, you'll be able to understand:*
 - *NM architecture*
 - *NM applications*
 - *NM platform*
 - *Guide on choosing a NM system*
 - *Introduction to Reference Model*



Generic NM Architecture

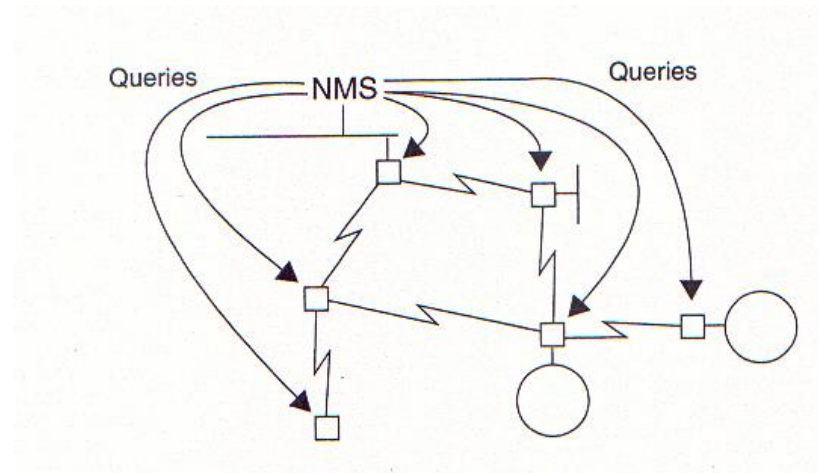


NM Architecture

- *NM platform can use various architectures to provide functionality.*
- *The three (3) most common NM architectures are:*
 - *Centralized*
 - *Hierarchical*
 - *Distributed*
- *There is no “best” architecture because each type has specific features that work well in certain environment.*
- *Each type of architecture has some advantages and disadvantages and to choose the architecture is according to the organizational structure.*

Centralized Architecture

- *It has the NM platform on one computer system, at a location that is responsible for all NM duties.*
- *This system uses a single centralized database and this system is backed up to another system at regular intervals.*
- *Even though it is centralized, it can allow access and can forward events to other consoles throughout the network.*



Centralized Architecture (cont'd...)

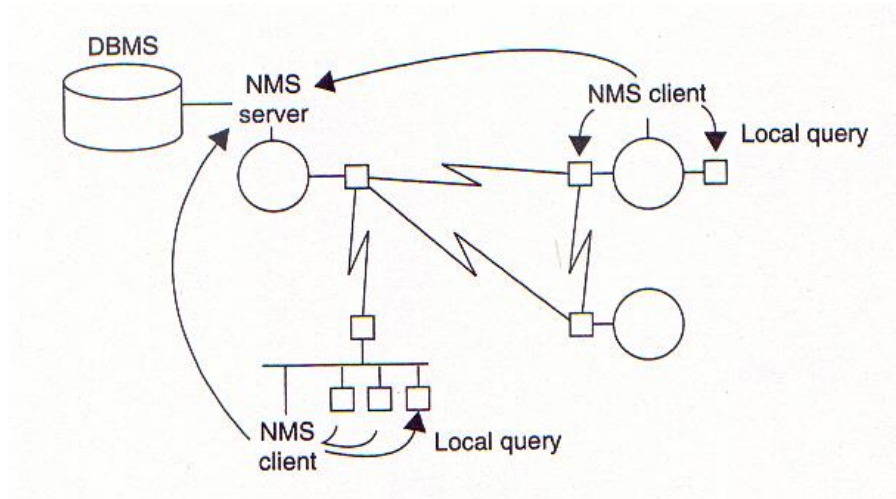
- *The single location of a centralized architecture is used:-*
 - *For all network alerts and events*
 - *For all network information*
 - *To access all mgmt applications*
- *NE can have a single location to view all network alerts and events, which is useful for troubleshooting and problem correlation.*
- *Having one place to access all of the NM application and information provides convenience, accessibility and security for NE.*

Centralized Architecture (cont'd...)

- *Having all NM functions depend on a single system, full backups should be maintained, ideally at another physical location.*
- *As network elements are added, it may be difficult and expensive to scale a single system to handle the necessary load.*
- *If connection to NM station fails, so all functions of NM will be lost.*
- *Example of centralized architecture: IBM Netview SNA.*

Hierarchical Architecture

- *It uses multiple systems, with one system acting as a central server and the others working as clients. (client-server)*
- *Some of the functions of the NM platform reside within the server, others run on the clients.*
- *The platform could use client/server database technology.*
- *The clients would not have separate database systems but would use the central server database accessed through the network.*
- *It will require backups for redundancy*



Hierarchical Architecture (cont'd...)

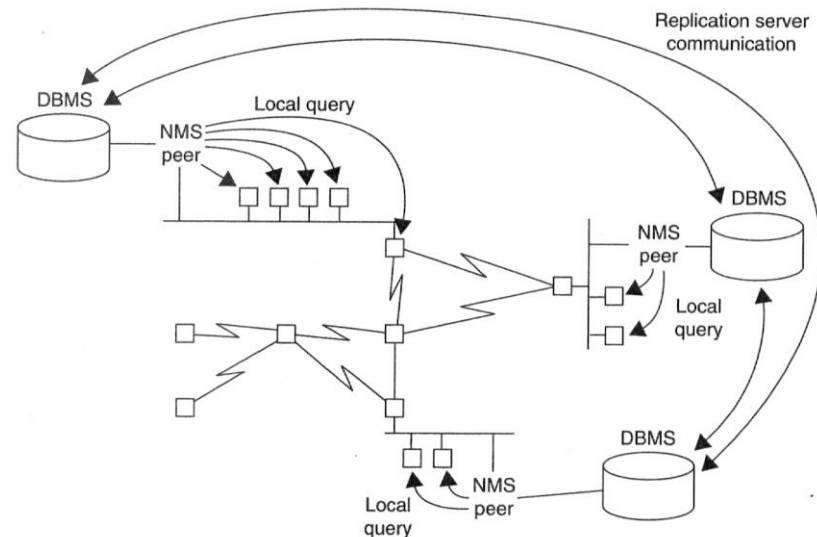
- *It has the features:*
 - *Not dependent on a single system*
 - *Distribution of NM tasks*
 - *Network monitoring distributed throughout the network*
 - *Centralized information storage*
- *It helps to alleviate one of the problems in a centralized approach by distributing NM tasks between the central system and the clients.*
- *NE can distribute network monitoring on the clients, saving valuable bandwidth resources throughout the network.*

Hierarchical Architecture (cont'd...)

- *Information gathering a bit more difficult and time consuming because it uses multiple systems to manage the network.*
- *The list of devices managed by each client needs to be logically predetermined and manually configured to avoid the both central system and client monitoring or polling the same device.*
- *I.e platform: SunConnect SunNet Manager, HP OpenView, IBM Netview/AIX, AT&T Star Sentry*

Distributed Architecture

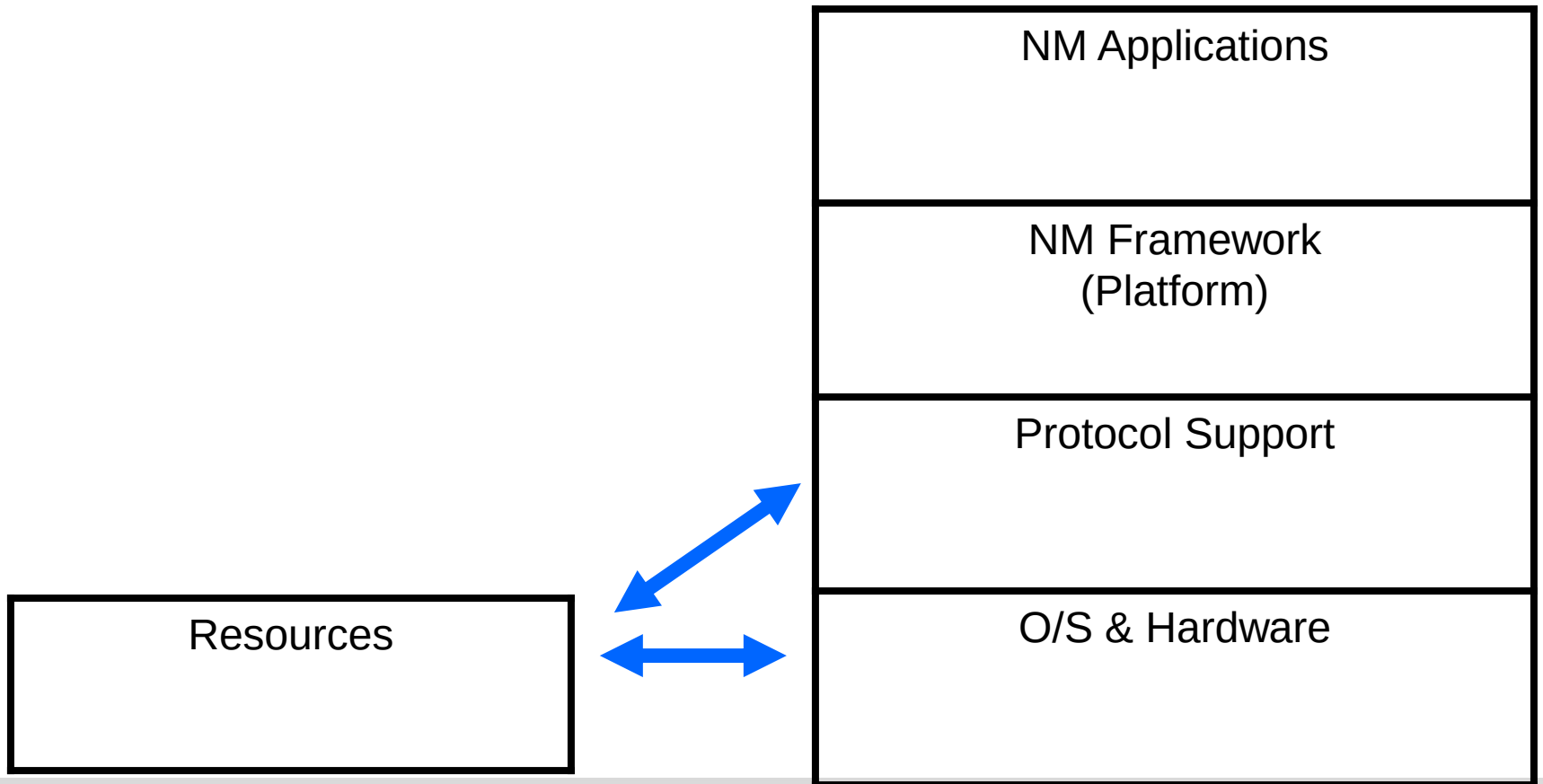
- *Combines the centralized and hierarchical approaches.*
- *It uses multiple peer platforms*
- *One platform is the leader of a set of peer NMS.*
- *Each individual peer platform can have a complete database for devices throughout the entire network which allows it to perform various tasks and to report the results back to a central system.*



Distributed Architecture (cont'd...)

- *Advantages of this architecture:-*
 - *Single location for all network information, alerts and events.*
 - *Single location to access all mgmt applications.*
 - *Not dependent on a single system*
 - *Distribution of NM tasks*
 - *Distribution of network monitoring throughout the network*
- *This architecture uses database replication server technology that keeps multiple databases on different systems completely synchronized.*

NMS Level



O/S & Hardware

- *O/S ~ DOS, Windows NT, OS/2 Warp, Novell Netware etc.*
- *Hardware ~ Computer, mainframe, server, switches, router etc..*

Resources

- *Separate with O/S and hardware*
- *PC ~ NIC, monitor, hard disks, software & applications.*

Protocol Support

- *Provide services:*
 - *Layers below application layer in OSI and Internet layer for TCP/IP (UDP = transport layer, IP = network layer)*
 - *SNMP, CMIP, CMOL (CMIP over LLC ~ logical link control)*

NM Framework (Platform)

- *Provide basis for NM applications.*
 - *Manager and agent functions*
 - *Database support*
 - *View and user interface support*
 - *Functions of NM*
- *Use by vendor to write applications for NM.*

NM Applications

- *Business Mgmt applications*
- *Easy-to-use view applications*
- *Fault identifications and fault diagnostic applications*
- *Performance-tuning applications*
 - *Sniffer Pro*
 - *Watchdog – H.A's by Professional Protocol*
 - *Tune Up by Olympus Software*
 - *EcoTOOLS by Compuware*

NM Concepts

- *NMS contain only protocols that convey information about network resources/elements from various agents in the system to manager or vice versa.*
- *There are many standards in NM:*
 - *OSI NM*
 - *Internet NM*
 - *IEEE NM*

Manager

- *Also known as managing process, NM station, Network Control Station, NM system.*
- *Responsible for activities such as:*
 - *5 functional areas of NM*
 - *Configuration mgmt: gather data*
 - *Directs the operations of the agents*

Agent

- *Also known as agent process or managed agent.*
- *Reports to the manager on the status of the MO and receives directions from the manager on actions it is to perform on these MO.*
- *It resides on workstation, PC, server, gateways, bridges, routers, switches and should be activated.*

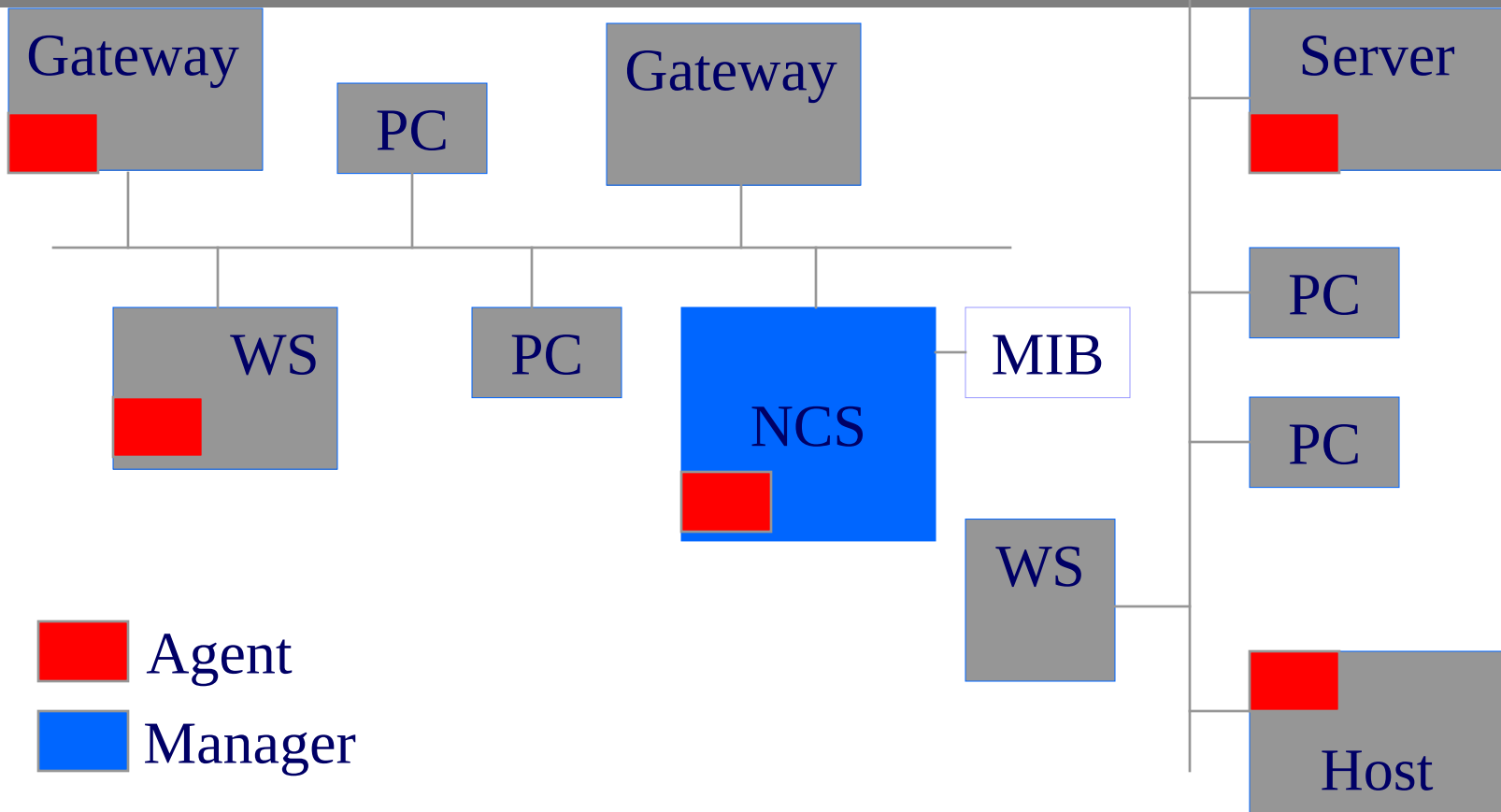
MIB

- *Management Information Base, also known as object library.*
- *Database used by both the agent and manager to determine the structure and content of management information (MO)*

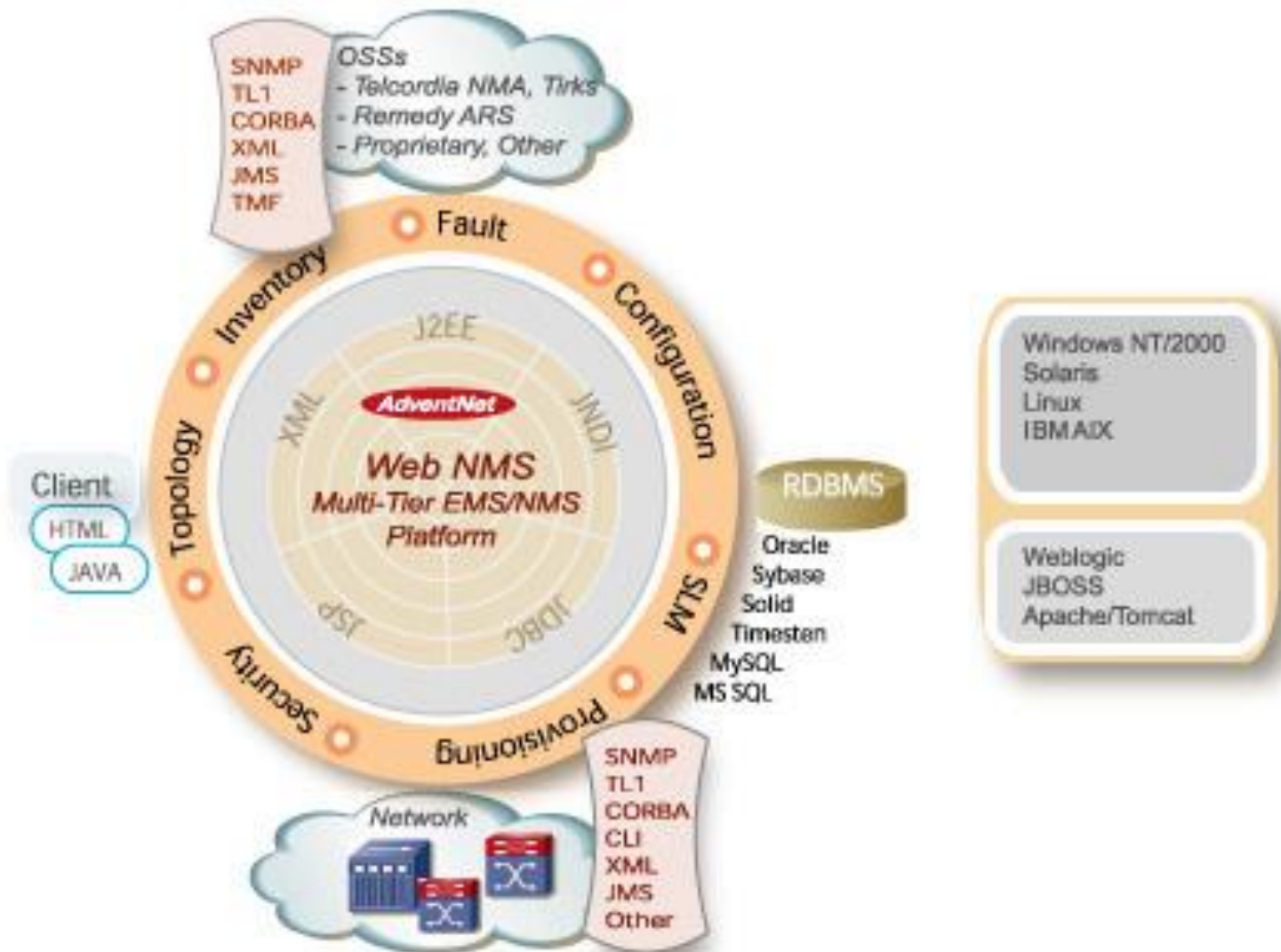
Managed Object (MO)

- *Also known managed network elements.*
- *The resources that are supervised and controlled by NM.*
- *It includes hardware and software components*
 - *physical media and connection (cables, ports), computer components (storage, processors), telecommunication hardware (modems, switches)*
 - *O/S software, Applications software, System software*

NM Components



Example: Web NMS



Mgmt Functions & Ref Models

- Serve as **conceptual** frameworks for organizing different tasks and functions that are part of network management.
- In reality the models are not literally followed – due to MS and support environ. – not the same.
- Use as a guidance and helps provide a sense of orientation.

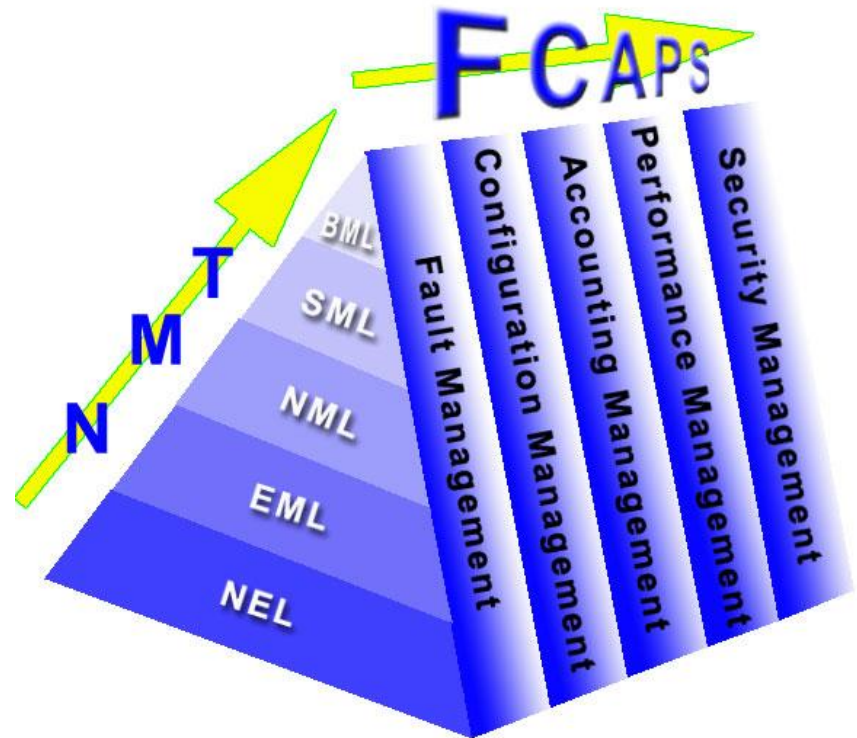
Mgmt Functions & Ref Models

- *It guides and helps in the following ways:*
 - *Easier to check a management system or operations support infrastructure for completeness.*
 - *Categories and group different functions, and identify which ones are closely related and belong together and which ones do not.*
 - *Identify scenarios and use cases that need to be collected, and to recognize interdependencies and interfaces between different tasks.*

FCAPS (*“eff-caps”*) model

- *One of the few reference model is FCAPS*
 - ***F** – Fault management*
 - ***C** – Configuration management*
 - ***A** – Accounting management*
 - ***P** – Performance management*
 - ***S** – Security management*
- *A part from Telecommunications Management Network (TMN)*
- *Will be touched in this course.*

TMN refined with FCAPS



AOM&P Model

- *Another reference model*
 - *A – Administration*
 - *O – Operations*
 - *M – Maintenance*
 - *P – Provisioning*
- *Large telecommunications service providers traditionally favor the AOM&P model.*

Mgmt Functions & Ref Models

- *The reference models are not the only one.*
- *NM can be organized in thousand of ways.*
- *Discussing reference models teaches important lessons regarding established ways to think about NM.*

Mgmt Functions & Ref Models

- *Points should be kept in mind while considering reference models:*
 - *Reference model is conceptual*
 - *Have specific constraints*
 - *Being able to slice up a problem space*



Thank you

